UMS UNIVERSITI MALAYSIA SABAH — FACULTY OF COMPUTING & INFORMATICS / FAKULTI KOMPUTERAN DAN INFORMATIK

BORANG DEKLARASI (BERKUMPULAN) / *DECLARATION FORM* (*GROUP*) UNTUK PENILAIAN MOD *ASYNCHRONOUS* / FOR ASYNCHRONOUS MODE ASSESSMENT

A. MAKLUMAT PELAJAR/*STUDENT'S INFORMATION*

Nama Penuh Ketua / Leader's Full Name: [Mengikut MyKad atau paspot / *As in MyKad or Passport*]	LIANA SYAFIERA BINTI LAI CHIU HIUNG
No. Matrik Ketua / Leader's *Matric No.*:	BI23110318
No. Telefon dan e-mel ketua / Leader's Phone No. and e-mail:	01154112092 and lianasyafierah@gmail.com
Nama Kursus / *Course Name*:	INTERNET OF THINGS
Kod Kursus / *Course Code*:	KK24602
Jenis Penilaian / *Assessment Type* :	☐ Tugasan / *Assignment* ☐ Kuiz / *Quiz* ☐ Lain-lain / *Others* : (Nyatakan / *Specify*) ..... ☐ Latihan Makmal / *Lab Exercise* / Projek / *Project*

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No. K.P. / Paspot Ketua: *Leader's I.C. no. / Passport:* 041120-12-0762

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D. SENARAI AHLI KUMPULAN / LIST OF GROUP MEMBERS

NO.	NAMA / NAME	NO. MATRIKS / MATRICS NUMBER	NO. KAD PENGENALAN / I.C. NUMBER	TANDATANGAN / SIGNATURE
1.	LIANA SYAFIERA BINTI LAI CHIU HIUNG	BI23110318	041120-12-0762	
2.	OLYN OLIVIA HERMAN	BI23110311	041024-12-0260	
3.	NURAINI MATNOR	BI23110295	040902-12-0922	
4.	NANDA RISTAH BINTI MUHD YUSUF	BI23110249	040126-12-0072	
5.	ABIGAIL ALAU R.SADOM	BI23110260	040319-12-0458	
6.				
7.				
8.				
9.				
10.				



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FACULTY OF COMPUTING AND INFORMATICS

KK24602 INTERNET OF THINGS

GROUP 3

SMART STREET LIGHT REPORT

SEMESTER 2 [2024/2025]

PREPARED BY

NO.	NAME	MATRIC NO.	MARK
1.	ABIGAIL ALAU R. SADOM	BI23110260	
2.	LIANA SYAFIERA BINTI LAI CHIU HIUNG	BI23110318	
3.	OLYN OLIVIA HERMAN	BI23110311	
4.	NURAINI MATNOR	BI23110295	
5.	NANDA RISTAH BINTI MOHD YUSUF	BI23110249	

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1.0 Project Proposal

1.1 Introduction

As urban populations grow, cities must adopt smarter, more sustainable infrastructure solutions. Street lighting, a vital yet energy-intensive municipal service, presents a significant opportunity for innovation. Our project presents a solar-powered smart streetlight system that combines IoT technology, renewable energy, and intelligent automation to create an efficient, eco-friendly lighting network. By integrating PIR motion sensors, LDR light sensors, ESP32-CAM for CCTV surveillance, and an ESP32 board, this system addresses energy waste, high costs, and security concerns while promoting environmental sustainability.

1.2 Problem statement

Despite its vital role in ensuring public safety and enhancing urban aesthetics, the current streetlight infrastructure is fraught with inefficiencies and shortcomings that hinder its effectiveness and sustainability. One of the most pressing issues is energy waste, as traditional street lights typically operate at full brightness from dusk until dawn, regardless of actual pedestrian or vehicular activity. This results in a significant and unnecessary drain on energy resources, particularly during periods of low foot traffic such as late at night. Closely tied to energy inefficiency are high operational costs. The constant use of electricity for lighting, coupled with the expenses associated with routine inspections, repairs, and replacements, places a considerable financial burden on municipal governments. These expenditures can consume a large portion of a city's public works or utilities budget, limiting the availability of funds for other critical urban services. Another widespread problem is poor maintenance management. In conventional systems, there is typically no automated way to detect lamp failures, meaning that broken or malfunctioning lights often go unnoticed until a citizen reports the issue. This reactive approach to maintenance leads to delayed repairs and extended periods of darkness, which can compromise public safety and erode public trust in local governance. Furthermore, the issue of light pollution has become increasingly prominent. Traditional streetlights, especially those that are poorly shielded or overly

bright, contribute to excessive night-time illumination. This disrupts natural ecosystems by confusing wildlife and interfering with their behavioral patterns. It also has adverse effects on human health, contributing to sleep disorders and disrupting circadian rhythms among urban residents.Finally, current systems often lack integrated security features. Most street lights operate in isolation without any capability for real-time surveillance or incident detection. This limits their potential role in enhancing urban security, such as monitoring for suspicious activity, aiding in crime prevention, or providing video evidence in the event of an incident.Together, these challenges underscore the urgent need for an upgraded, intelligent street lighting solution that can adapt to environmental conditions, conserve energy, reduce costs, improve maintenance efficiency, minimize light pollution, and enhance public safety through advanced monitoring capabilities.

1.3 Proposed IoT solution

In response to the inefficiencies and challenges posed by conventional street lighting infrastructure, we propose the development and deployment of a Smart Street light System that leverages modern electronics, renewable energy, and intelligent automation to create a more efficient, responsive, and sustainable urban lighting network. Central to this system is the use of an ESP32, a cost-effective and versatile microcontroller that serves as the brain of the operation, coordinating data from the sensor and managing light behavior accordingly.

The proposed system employs Passive Infrared (PIR) sensors to detect the presence of pedestrians or vehicles, allowing lights to automatically adjust their brightness based on real-time activity. This motion-sensitive functionality ensures that energy is used only when necessary, significantly reducing waste. In addition, Light Dependent Resistor (LDR) is incorporated to monitor ambient light levels, enabling the system to intelligently differentiate between natural and artificial light requirements. As a result, the street lights automatically turn off during daylight or adequately lit conditions, thereby enhancing energy efficiency even further.

To promote sustainability and reduce dependency on grid electricity, the system is powered primarily by solar panels, which store energy during the day and utilize it to power the lights at night. This shift toward renewable energy not only lowers electricity costs but also reduces the carbon footprint of municipal lighting services. To enhance urban safety and provide basic visual monitoring of public spaces, the proposed system includes the integration of a ESP32-CAM module, which functions as a compact and cost-effective CCTV camera. This module allows for real-time video streaming, providing a visual record of activity in the vicinity of each streetlight. The presence of the ESP32-CAM adds a valuable layer of passive surveillance, potentially deterring criminal activity and offering visual evidence in the event of an incident. The camera's feed can be accessed locally or transmitted over a network, depending on connectivity options available, making it a practical solution for low-cost, scalable deployment in urban or semi-urban environments.

In summary, this Smart Street Light System presents a comprehensive solution that not only addresses current limitations but also supports the broader goals of smart city development, energy conservation, and public safety enhancement.

1.4 How the Smart Street light Works

The operation of the smart street light is based on a coordinated interaction between sensors, a microcontroller, a power system, and optional monitoring components. Designed for energy efficiency and intelligent automation, the streetlight functions as follows.

During the daytime, solar panels mounted on or near the street light harness sunlight and convert it into electrical energy. This energy is stored in rechargeable batteries, which are used to power the system at night. This renewable energy approach significantly reduces reliance on the electrical grid and contributes to environmental sustainability.

As evening approaches and ambient light levels drop, the Light Dependent Resistor (LDR) sensor detects the decreasing natural light. The system interprets this data using the ESP32, which acts as the main controller. Once the ambient light falls below a specific threshold, the controller activates the LED light using Pulse Width Modulation (PWM) to regulate its brightness.

When no motion is detected, the light remains in a dimmed or low-power state, conserving energy. However, when the PIR (Passive Infrared) motion sensor detects movement from a pedestrian or vehicle, it sends a signal to the ESP32. The microcontroller responds by increasing the brightness of the LED light to ensure clear visibility for people in the area.

The system also includes a basic surveillance feature using an ESP32-CAM module. This small, Wi-Fi-enabled camera can stream video over a wireless connection. When motion is detected, the ESP32-CAM can be configured to begin recording or transmitting footage, providing an added layer of security and visual monitoring.

Overall, the smart streetlight operates autonomously by responding to real-time environmental and activity data. It provides reliable illumination only when needed, maximizes energy efficiency through solar power and sensor-based control, and supports improved public safety with integrated surveillance features.

1.5 Expected impacts

By introducing this Smart Streetlight System, we're not just upgrading lights but also reshaping how our cities use energy, respond to people's needs, and create safer public spaces. The PIR sensor detects movement and when there's no activity, the light dims to save energy. The LDR sensor, on the other hand, keeps an eye on natural light levels. If it's still bright outside, the streetlight stays off or dimmed, which means we're not wasting electricity when sunlight is doing the job for us. By combining both sensors, the system delivers the right amount of light at the right time. This smart approach can lead to up to 70% reduction in energy costs, which is a huge relief for tight city budgets.

The fact that the system runs on solar power makes it even more sustainable. During the day, solar panels charge the batteries, and at night, that stored energy powers the lights. This means less dependence on the grid, fewer emissions, and a meaningful step toward reducing the city's carbon footprint.

Safety also gets a boost. Lights automatically brighten when someone approaches, making people feel more secure at night. And the system includes a basic ESP32-CAM module, a small camera that connects to Wi-Fi and can record or stream video when activated by the system. It is a helpful layer of visibility that can deter unwanted activity and support community safety.

2.0 Project Methodology

The development of the Smart Street Light followed a structured methodology to ensure efficiency, functionality and reliability. The process was divided into distinct phases : research and planning, hardware assembly, software development, integration and testing. Initial research focused on identifying the limitations of conventional street lighting systems and exploring IoT-based solutions. Key components such as the ESP32 module, PIR sensor, LDR sensor, ESP32-CAM and solar panel were selected based on their compatibility, cost-effectiveness, and performance. A system design was drafted to outline the interaction between inputs (sensors), processing (microcontroller), and outputs (LEDs, Telegram alerts).

Moving on to the hardware assembly phase, the hardware implementation began with assembling the core components. The ESP32 module served as the central controller, connected to the PIR motion sensor for detecting movement and the LDR sensor for monitoring ambient light. The ESP32-CAM was integrated for surveillance, while the solar panel and Li-ion battery ensured sustainable power. Jumper wires and a breadboard facilitated temporary circuit connections during prototyping. Voltage regulators like the MT3608 boost converter stabilized power delivery to components requiring higher voltages. In the software development phase, the system's intelligence was programmed using the Arduino IDE. Key libraries such as WiFi.h, UniversalTelegramBot.h, and ArduinoJson.h enabled Wi-Fi connectivity, Telegram integration, and JSON data parsing. The control logic was designed to process sensor data in real time which is the LDR determined daylight conditions, while the PIR triggered LED brightness adjustments upon detecting motion. Telegram alerts were configured to notify users of system status, errors, or motion events. During the integration phase, hardware and software components were merged to create a cohesive system. Sensor data was validated to ensure accurate responses, and the ESP32-CAM was tested for live video streaming via a local hotspot. The solar power system was optimized to charge the battery efficiently and sustain operation during nighttime. The final phase is testing and evaluation. During this phase, rigorous testing was conducted to evaluate performance under various conditions. Sensor accuracy, response times, and energy efficiency were measured. The Telegram bot's reliability in sending alerts was verified, and the ESP32-CAM's video quality was assessed. Adjustments were made to refine the system's responsiveness and stability.

This systematic approach ensured the Smart Street Light system met its objective of energy efficiency, automation and enhanced security while adhering to IoT best practices.

3.0 System Design

3.1 IPO Diagram

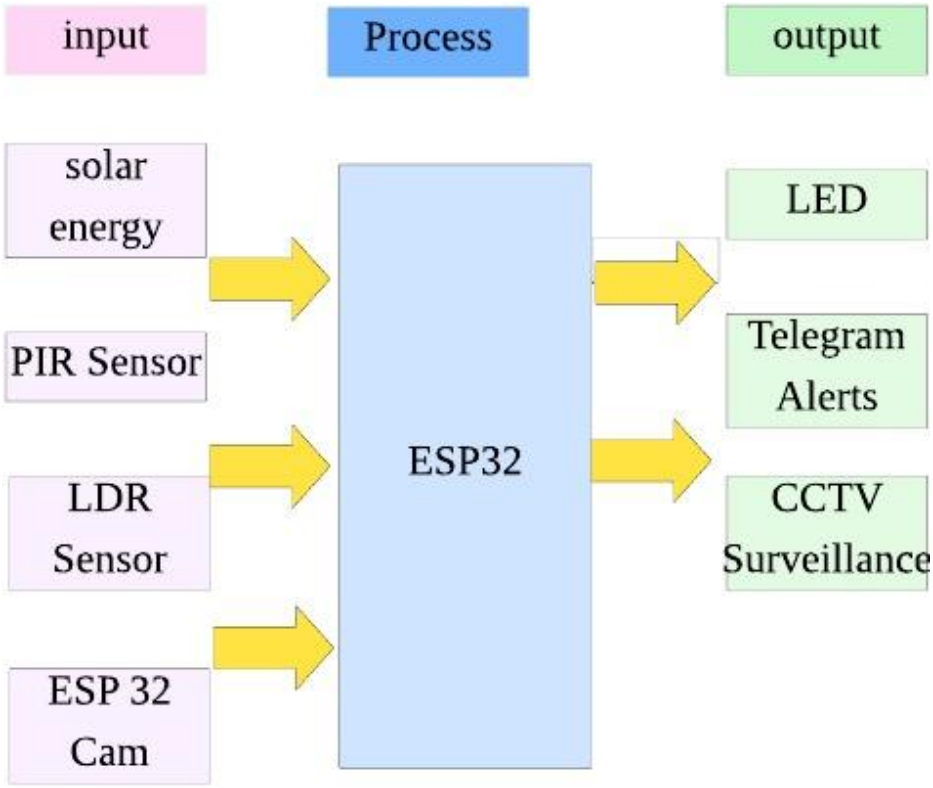


Figure 1 IPO diagram

The solar monitoring system outlined in the diagram is designed to automate environmental sensing and alerting using a combination of input sensors, a central processor (ESP32), and output components. This system integrates Passive Infrared (PIR) and light-dependent resistor (LDR) sensor to detect motion and ambient light levels, respectively, and communicates status updates via Wi-Fi to a Telegram bot for remote notifications. The system begins with data acquisition through four primary input sources. The first input is a solar panel. The solar panel harvests energy to power the ESP32 and connected sensors to promote energy efficiency and off-grid operation. Secondly, the PIR Sensor. The passive infrared sensor detects movement within its range. This could serve security purposes or trigger energy saving actions. The third input is the LDR Sensor. The light-dependent resistor measures ambient brightness. This data helps assess whether solar panels are receiving adequate sunlight or if artificial

lighting is needed. Fourthly, the ESP32-CAM module. The ESP32 CAM captures visual data when motion is detected and also supports real-time video streaming.

Next is the Central Processing which the ESP32 microcontroller acts as the system's brain, performing data processing and wifi communication. The ESP32 reads digital signals from the PIR and LDR sensors. For example, If the PIR sensor detects motion, the ESP32 activates the LED. If the LDR reports low light levels, the ESP32 may trigger an alert. The ESP32 also connects to the internet via Wi-Fi to send real-time updates to a Telegram bot.

Last but not least, the Output. The system converts data into clear output. One of the outputs is the LED Indicator. It provides immediate visual feedback. For instance, Steady ON means Motion detected (PIR sensor) and Blinking means Low light levels (LDR sensor). Next output is the Telegram Alerts. Errors like sensor disconnections are pushed to the user via Telegram. Finally is the CCTV Surveillance where the footage were stored and recorded successfully.

3.2 System Implementation Workflow

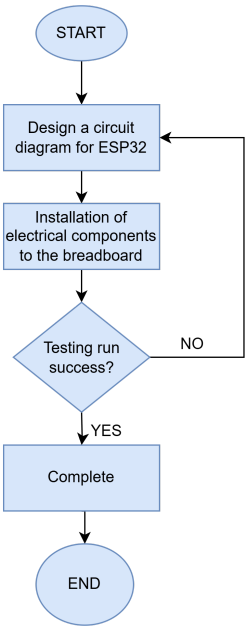


Figure 2 System Implementation Workflow

The system implementation follows a structured workflow to ensure seamless integration of hardware and software components. The process begins with designing the circuit diagram for the ESP32 microcontroller which serves as the central control unit of the Smart Street Light. This step involves mapping connections between the ESP32, sensors (PIR and LDR), power modules and output components (LEDs and ESP32-CAM) to ensure compatibility and functionality.

Once the circuit design is finalized, the next phase involves the physical installation of electrical components onto a breadboard for prototyping. This stage allows for preliminary testing of connections and immediate troubleshooting of any wiring or power issues. Key components, such as the PIR motion sensor, LDR light sensor, and voltage regulators, are carefully connected to the ESP32 to validate their interactions. After assembly, the system undergoes rigorous testing to verify sensor responsiveness, power stability, and communication with the Telegram bot. If any component fails to perform as expected, the process loops back to redesign or reinstallation until all elements operate correctly. Only after successful validation is the system deemed complete and ready for deployment. This iterative approach ensures reliability and adherence to the project's objectives of energy efficiency and automation.

3.3 Power and Data Flow Diagram

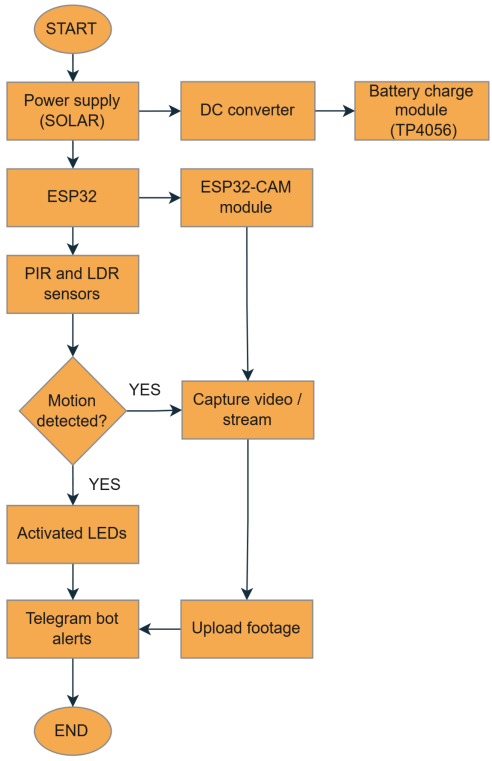


Figure 3 Power and Data Flow Diagram

The power and data flow diagram illustrates the hierarchical structure of energy distribution and information processing within the Smart Street Light System. The system is primarily powered by a solar panel, which harvests renewable energy during daylight hours. This energy passes through a DC converter to stabilize the voltage before being stored in a rechargeable Li-ion battery via a TP4056 charging module. The battery then supplies power to the ESP32 microcontroller and peripheral components, including the ESP32-CAM module, ensuring uninterrupted operation during nighttime or low-light conditions. Sensor data is collected through the PIR motion sensor that detects human presence and ambient light levels. These inputs are processed by the ESP32 which executes pre-programmed logic to control the brightness of the LED street light.

For instance, upon detecting motion in low-light conditions, the ESP32 triggers the LEDs to illuminate at full intensity. The ESP32-CAM module enhances the system's functionality by providing real-time visual surveillance. When the PIR sensor detects motion, the ESP32 not only activates the LEDs but also signals the ESP32-CAM to capture images or stream video. The camera module operates independently yet synchronously with the main ESP32, leveraging Wi-Fi to transmit footage to a local server or cloud storage. This dual-microcontroller approach ensures efficient resource allocation where the ESP32 handles sensor and lighting control while the ESP32-CAM manages video processing without overburdening the primary system.

Additionally, the system integrates IoT capabilities through a Telegram bot, enabling remote monitoring and alerts. The ESP32 transmits real-time data such as motion detection logs, camera activation events, or system errors to the Telegram platform, providing administrators with instant updates. This bidirectional flow of power and data, now augmented with visual surveillance, ensures a comprehensive, responsive, and sustainable street lighting solution.

3.4 Circuit Implementation

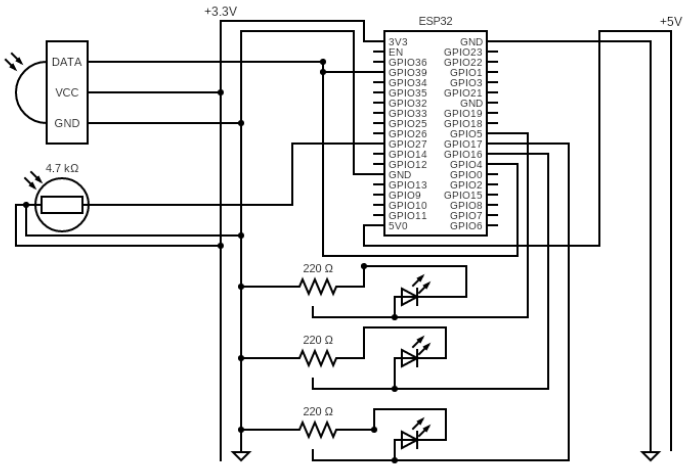


Figure 4 Circuit Implementation

The **ESP32 microcontroller** serves as the central control unit in the Smart Street Light system, managing inputs from sensors and controlling outputs such as LEDs. A **PIR motion sensor** is connected to **GPIO27**, allowing the ESP32 to detect human presence or movement in the area. When motion is sensed, this input triggers the microcontroller to activate the street lights or initiate a response from the camera module.To monitor ambient light levels, a **photoresistor (LDR)** is connected to two pins on the ESP32. Its **analog output** is wired to **GPIO39**, enabling the ESP32 to measure light intensity on a continuous scale, while its **digital output** is connected to **GPIO4** for threshold-based decisions, such as turning the lights on when darkness is detected.

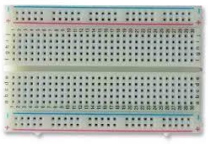
The system features three **white LEDs** that provide illumination, each connected to a separate GPIO pin for individual control: **LED1 to GPIO5**, **LED2 to GPIO16**, and **LED3 to GPIO17**. Each LED includes a **220Ω resistor** in series to limit current and protect the LEDs. These LEDs are activated or deactivated based on the combined input from the PIR sensor and LDR.The ESP32 itself is powered via its **5V VIN pin**, which receives a stable voltage from the **MT3608 boost converter**. This converter increases the 3.7V output from the lithium-ion battery to 5V, ensuring reliable operation of the microcontroller and its connected components. Overall, the ESP32 efficiently coordinates sensor data processing and lighting control, making it the core of the smart street lighting system.

4.0 Implementation

4.1 Hardware Component

Component	Function	Specifications
ESP32 Board 	The microcontroller serves as the central processing unit that coordinates sensor data acquisition, lighting control logic and wireless communication with the Telegram API.	• Dual-core 240MHz • Support Wi-Fi and Bluetooth • 3.3V logic
ESP32 CAM 	This imaging module works as a CCTV monitoring camera that captures and streams video footage when activated by the system.	• OV2640 camera • 5V input
PIR Motion Sensor 	The passive infrared sensor detects human presence by monitoring changes in thermal radiation patterns within its effective range.	• 3.3V to 5V input • Range: up to 4m
Photoresistor (LDR) 	This light-sensitive component measures ambient illumination levels to determine appropriate lighting activation periods.	• Analog (0-3.3V) • Resistance: 10kΩ for dark to 1kΩ for light
White LED	These illumination elements	• 5mm

	provide the actual lighting output with intensity controlled through microcontroller signals.	• 3.2V
3.7V Li-ion Battery 	The rechargeable power source enables autonomous system operation through its energy storage capacity.	• 18650 cell • Capacity: 2000mAh to 3500mAh
TP4056 Charging Module 	This charging module manages safe power input from both solar and USB sources to maintain battery charge.	• Input: 5V from solar or micro-USB
Solar Panel 	The photovoltaic component harvests environmental solar energy to sustain long-term system operation.	• 1W to 5W output • 5V for open-circuit voltage
MT3608 Boost Convertor 	This voltage regulation component ensures stable 5V power delivery to modules requiring higher operating voltages.	• Input: 2V • Output: 5V

Jumper Wires 	These conductive connections establish the necessary electrical pathways between system components.	• Male-to-male • Female-to-female • Female-to-male
Breadboard 	The prototyping platform facilitates temporary circuit construction during system development and testing.	• 400 tie-points
220Ω Resistor 	These current-limiting components protect sensitive elements from excessive electrical current.	• 1W power rating

4.2 Hardware Implementation

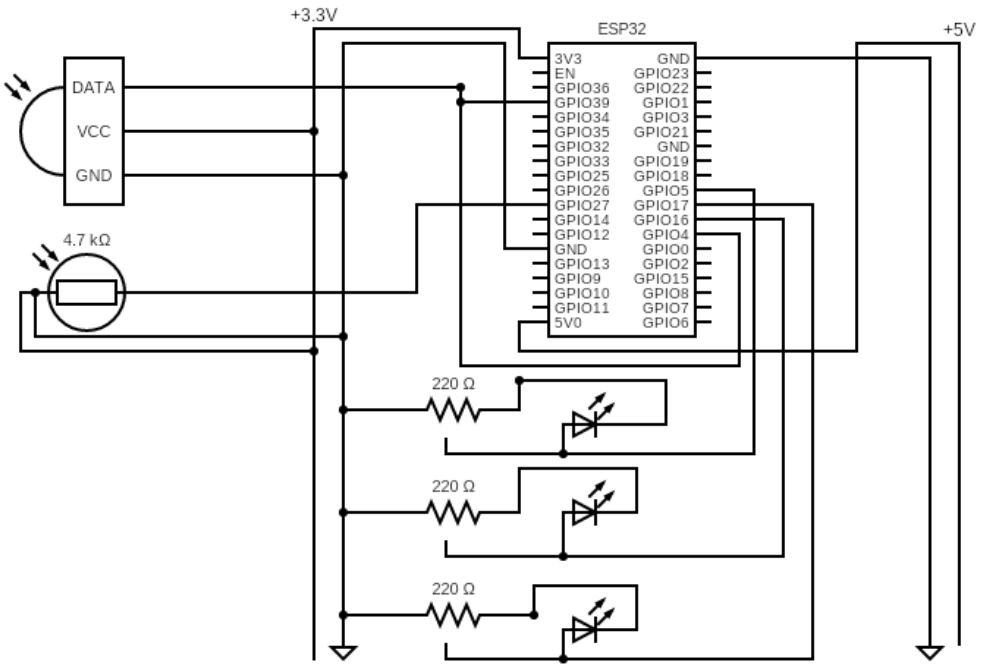
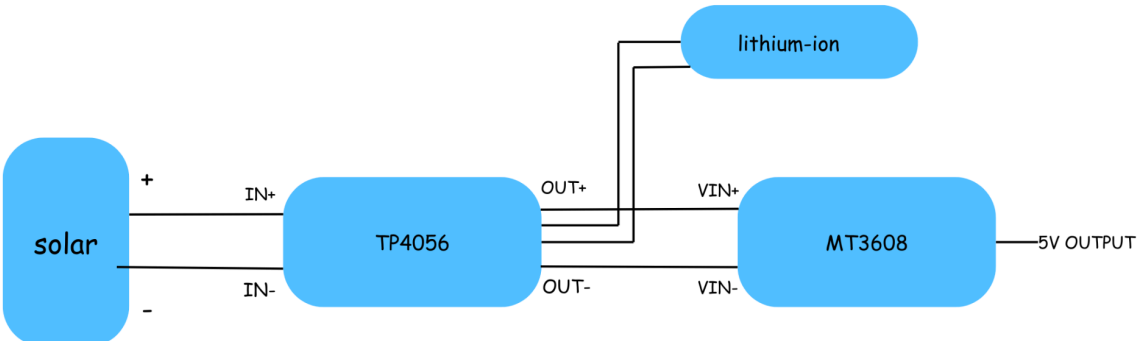


Figure 5 Circuit Implementation

ESP32 Microcontroller	
Function	Central processing unit
Connections	● LDR analog output connected to analog input GPIO39. ● LDR digital output connected to a digital input GPIO4. ● PIR sensor connected to a digital input GPIO27. ● White LED 1 connected to GPIO5, LED 2 connected to GPIO16 and LED 3 connected to GPIO17 via a 220Ω resistor. ● Powered through the 5V output from the MT3608 boost converter.
ESP32-CAM	
Function	Surveillance camera
Connections	● Powered via 5V from the USB laptop since the laptop gives a stable power and can be used to upload a code to view the camera output. ● The ESP32-CAM turns on and creates a Wi-Fi network and connects the Wi-Fi to the phone or laptop. Then, open a browser and go to the special address which is *192.168.4.1* to see what the camera is showing.

PIR Motion Sensor	
Function	Detects motion
Connections	● Output connected to a digital input of the ESP32 (GPIO27). ● VCC to 3.3V and GND to system ground on the breadboard.
Photoresistor (LDR)	
Functions	Detects light intensity
Connections	● Analog output connected to analog input GPIO39. ● Digital output connected to digital input GPIO04. ● GND connected to ground on the breadboard (left negative rail of breadboard). ● VCC to 3.3V (left positive rail of breadboard)
White LED	
Function	Provides illumination.
Connections	● LED1 ○ **Anode** : connected to GPIO5 ○ **Cathode** : connected to ground rail of breadboard through 220Ω resistor ● LED2 ○ **Anode** : connected to GPIO16 ○ **Cathode** : connected to ground rail of breadboard through 220Ω resistor ● LED3 ○ **Anode** : connected to GPIO17 ○ **Cathode** : connected to ground rail of breadboard through 220Ω resistor
Breadboard and Jumper Wires	
● Use for prototyping all the above connections in a modular and testable format.	



Li-ion Battery (3.7V, 18650)	
Functions	Main power source
Connections	• Positive terminal to B+ on TP4056 • Negative terminal to B- on TP4056
TP4056 Charging Module	
Function	Manages battery charging
Connections	• Solar panel (positive and negative) connected to IN+ and IN- of TP4056. • OUT+ and OUT- connected to VIN+ and VIN-
Solar Panel (5V)	
Function	Recharges battery
Connections	• Connected to IN+ and IN- of TP4056 • Mounted for optimal sunlight exposure
MT3608 Boost Converter	
Function	Boost 3.7V to 5V
Connections	• Input from TP4056 OUT+and OUT- • Output USB 5V connected to ESP32 microcontroller

4.3 Software Implementation

4.3.1 Arduino IDE Configuration

The Smart Street Light system was developed using the Arduino Integrated Development Environment (IDE), which offers a user-friendly and flexible platform for programming embedded systems. The ESP32 Dev Module was chosen as the microcontroller board due to its powerful processing capabilities and integrated Wi-Fi, making it suitable for Internet of Things (IoT) applications. To support the functionality of the system, several essential libraries were included in the project. These include WiFi.h for enabling wireless network connectivity, WiFiClientSecure.h for handling secure HTTPS communication, UniversalTelegramBot.h for enabling interaction with Telegram's Bot API, and ArduinoJson.h for parsing JSON-formatted data. The system initializes by connecting to a predefined Wi-Fi network using stored credentials of SSID and password. For simplicity and ease of communication, SSL certificate verification is disabled using client.setInsecure(), which allows encrypted communication without validating the certificate. Serial communication is also initiated using Serial.begin(), enabling real-time monitoring of the system's status, such as network connectivity and sensor activity. This provides essential feedback during both development and deployment, helping to ensure stable and reliable operation of the system.

```
#include <WiFi.h>
#include <WiFiClientSecure.h>
#include <UniversalTelegramBot.h>
#include <ArduinoJson.h>
#include "esp_camera.h"

// WiFi credentials
const char* ssid = "Unifi";
const char* password = "miloping20";

// Telegram bot token and chat ID
const char* BOT_TOKEN = "7697063713:AAG1SNrCz2rx1j2OFOyPCvRHE0LiFzjneWo";
const char* CHAT_ID = "1014727422";

// Telegram setup
WiFiClientSecure client;
UniversalTelegramBot bot(BOT_TOKEN, client);
```

Figure 6 Smart Street Light Code Snippet

4.3.2 Telegram Bot Integration

To enable remote monitoring and control, the Smart Street Light system is integrated with Telegram through a custom bot. This bot was created using Telegram's BotFather tool, which generates a unique Bot Token (BOT_TOKEN) required for authentication. In addition, the Chat ID (CHAT_ID) of the intended recipient is specified to direct all communication to the correct user. Upon successful Wi-Fi connection, the ESP32 sends an automatic startup message " Smart Street Light is now online" to the specified Telegram chat, confirming that the system is operational. The UniversalTelegramBot library is utilized to process incoming Telegram messages and execute corresponding commands. For instance, when a user sends the /status command, the system responds with current information regarding sensor status or LED conditions. To prevent overloading the Telegram API and to enhance message efficiency, a cooldown interval of five seconds is implemented using the millis() function. This delay ensures that alerts, such as motion detection or environmental changes are sent at controlled intervals and preventing spam while maintaining responsive communication.

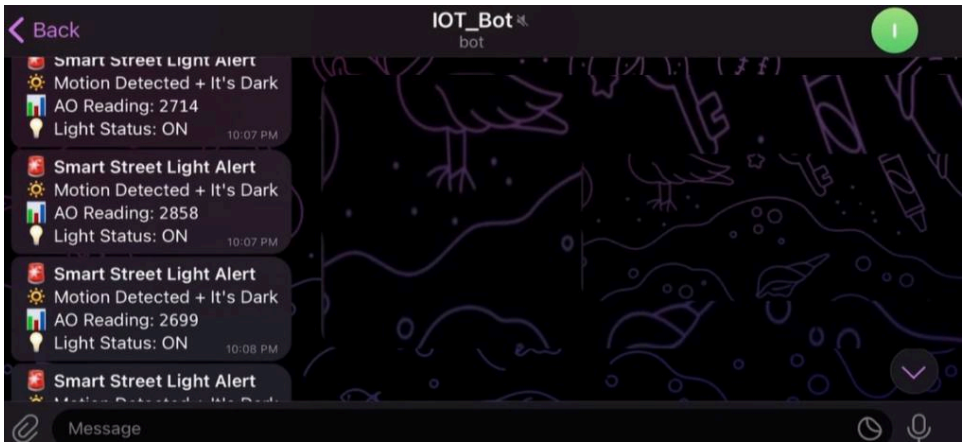


Figure 7 Telegram Bot

4.3.3 Sensor Integration and Control Logic

The automated behavior of the Smart Street Light system relies on the integration of two key sensors which are a Photoresistor Light Dependent Resistor (LDR) and a Passive Infrared (PIR) motion sensor. The LDR is used to detect ambient light conditions. It is connected to both a digital pin (LDR_DO_PIN) to indicate the presence or absence of light became HIGH when dark, and an analog pin (LDR_AO_PIN) to provide detailed intensity readings. The PIR sensor (PIR_PIN) detects human movement by outputting a HIGH signal when motion is present. The system continuously reads values from these sensors in the loop() function. If the environment is detected as dark and motion is simultaneously detected, the system turns on three LEDs (LED1_PIN, LED2_PIN, and LED3_PIN) to illuminate the area. During this condition, and only if the cooldown period has elapsed, a notification is sent via Telegram containing sensor readings and the status of the lights. If either condition is not met either it is not dark or there is no motion the LEDs remain off. All sensor readings and system actions are printed to the Serial Monitor to aid in debugging and ensure transparency in the system's decision-making process.

```
23  // Pin Definitions
24  #define PIR_PIN 27 // PIR sensor
25  #define LDR_DO_PIN 4 // LDR digital pin
26  #define LDR_AO_PIN 39 // LDR analog pin
27  #define LED1_PIN 5 // LED1
28  #define LED2_PIN 16 // LED2
29  #define LED3_PIN 17 // LED3
30
31  void setup() {
32  Serial.begin(115200);
33  // Initialize sensor and LED pins
34  pinMode(PIR_PIN, INPUT);
35  pinMode(LDR_DO_PIN, INPUT);
36  pinMode(LED1_PIN, OUTPUT);
37  pinMode(LED2_PIN, OUTPUT);
38  pinMode(LED3_PIN, OUTPUT);
39  analogSetAttenuation(ADC_11db);
40  digitalWrite(LED1_PIN, LOW);
41  digitalWrite(LED2_PIN, LOW);
42  digitalWrite(LED3_PIN, LOW);
```

Figure 8 Sensor Integration Code Snippet

4.3.4 ESP32 CAM Integration

An enhancement to the Smart Street Light system involves the integration of an ESP32-CAM module to provide surveillance capabilities through video streaming. This functionality is implemented in a separate .ino file, which configures the ESP32-CAM to operate as a standalone hotspot under the SSID "ESP32-CAM-Hotspot", with a default password of "12345678". Upon booting, the module initializes the onboard camera with predefined settings optimized for image quality and performance, such as JPEG format and UXGA frame size, depending on available PSRAM. The camera configuration ensures efficient image capture suitable for streaming purposes. After successful initialization, the module prints its local IP address to the Serial Monitor, allowing users to access the live video feed through a web browser connected to the hotspot. This addition enhances the system's functionality by offering real-time visual monitoring and increased security coverage.

```
97  // Start WiFi hotspot instead of connecting to external WiFi
98  const char* ap_ssid = "ESP32-CAM-Hotspot";
99  const char* ap_password = "12345678";
100 WiFi.softAP(ap_ssid, ap_password);
101 IPAddress IP = WiFi.softAPIP();
102 Serial.print("Hotspot IP address: ");
103 Serial.println(IP);
104 startCameraServer();
105 Serial.println("Camera Ready! Connect to the WiFi hotspot and open the IP above.");
```

Figure 9 ESP32 CAM Integration Code Snippet

5.0 Testing and Evaluation

5.1 Testing Methodology

The Smart Street Light system underwent rigorous testing to ensure functionality, reliability, and efficiency. The evaluation process was divided into three key phases:

a. Unit Testing – Individual components were tested in isolation.
b. Integration Testing – The interaction between hardware and software components was verified.
c. Field Testing – The complete system was evaluated under real-world conditions.

5.1.1 Unit Testing

Component	Test Case	Expected Result	Actual Result	Pass/Fail
ESP32 Board	Power-up and Wi-Fi connection	Boots successfully, connects to Wi-Fi	Connected toWi-Fi as expected	Pass
PIR Sensor	Motion detection in different ranges	Detects movement	Detected motion within specified range	Pass
LDR Sensor	Response to varying light conditions	Outputs correct analog/digital values	Adjusted correctly to light changes	Pass
ESP32-CAM	Video streaming	Stable live feed, clear image quality	Functional with minor latency	Pass

Solar Panel & Battery	Charging under sunlight, power retention	Battery charges fully, sustains system	Charged efficiently, powered system	Pass
Telegram Bot	Command response & alert notifications	Sends/receives messages without delay	Worked with slight delay (~2 sec)	Pass (Minor lag)

5.1.2 Integration Testing

- Motion-Activated Lighting:
  - Test: Wave hand in front of the PIR sensor in low-light conditions.
  - Result: LEDs turned on immediately and dimmed after motion ceased.
  - Issue: Slight delay (~1 sec) in response time.
- Automatic Daylight Shutoff:
  - Test: Covered/uncovered LDR to simulate day/night transitions.
  - Result: Lights turned off in bright conditions and reactivated in darkness.
- Surveillance Activation:
  - Test: Triggered PIR sensor to activate ESP32-CAM.
  - Result: Camera started streaming, footage was accessible via local IP.
- Telegram Alerts:
  - Test: Simulated motion events and system errors.
  - Result: Notifications were sent successfully but occasionally delayed.

5.1.3 Field Testing

Parameter	Observation	Conclusion
Energy Efficiency	Solar panel charged battery sufficiently	System ran autonomously overnight
False Triggers	Occasional false PIR activations such as wind	Improved filtering needed
Camera Performance	Video quality degraded in low light	Requires IR or low-light enhancement
Wi-Fi Stability	Occasional disconnections in weak signal areas	Needs better antenna or mesh network

5.2 Performance Evaluation

5.2.1 Energy Savings

Baseline Comparison:

In a conventional lighting system, streetlights remain continuously active throughout the night, regardless of human presence or ambient light levels. In the context of this prototype, the system uses three 5mm white LEDs, each with an approximate power consumption of **0.06W**, resulting in a total continuous consumption of:

$$\textbf{Total power} = 0.06W \times 3 = 0.18W$$

Assuming 12 hours of continuous night-time operation, the total energy usage in a traditional system would be:

$$0.18W \times 12h = 2.16Wh$$

Smart System Behavior:

The smart lighting system activates the LEDs **only when two conditions are met simultaneously**: it is dark (as detected by the LDR sensor), and motion is detected (via the PIR sensor). This significantly reduces unnecessary energy use. Based on observed usage during field testing, it is estimated that motion events requiring full LED activation occur approximately **20% of the night-time duration**. Thus, the estimated energy consumption of the smart system is:

$$0.18W \times (12h \times 20\%) = 0.18W \times 2.4h = 0.432Wh$$

Estimated Energy Savings:

$$\textbf{Energy Savings} = \frac{2.16Wh - 0.432Wh}{2.16Wh} \times 100\% \approx 80\%$$

Conclusion:

Although the absolute power values are low due to the scale of the prototype, the implementation demonstrates a clear potential for energy efficiency. The smart system achieves an approximate **80% reduction in energy usage** by utilizing motion and ambient light sensors for conditional activation. This concept can be scaled to real-world applications involving high-power street lighting, leading to substantial long-term savings and environmental benefits.

5.2.2 Response Time Analysis

Function	Average Response Time	Acceptable Threshold	Verification
PIR Activation	0.8 sec	≤1 sec	Within limits
LDR Adjustment	1.2 sec	≤2 sec	Within limits
Telegram Alert Delay	2.5 sec	≤3 sec	Needs optimization

5.2.3 Reliability & Durability

- Battery Life: Lasted 12+ hours on a full charge.
- Weather Resistance: Performed well in light rain but requires waterproof casing for long-term use.
- Component Longevity: No overheating issues observed during testing.

5.3 Limitations & Improvements

5.3.1 Identified Limitations

1. PIR Sensor Sensitivity: False triggers due to animals or wind.
2. ESP32-CAM Limitations: Poor night vision and limited resolution.
3. Wi-Fi Dependency: Requires stable network for remote monitoring.
4. Solar Dependency: Reduced efficiency on cloudy days.

5.3.2 Proposed Improvements

- Enhanced Motion Detection: Use microwave radar sensors for better accuracy.
- Low-Light Camera Upgrade: Integrate PIR LEDs for night vision.
- Mesh Networking: Implement LoRaWAN for wider coverage in remote areas.
- Energy Backup: Add dual power sources (solar + grid) for reliability.

5.4 Conclusion

The Smart Street Light system successfully demonstrated energy efficiency, automation, and security enhancements compared to traditional street lighting. While minor optimizations are needed, the system proves viable for smart city applications. Future iterations will focus on scalability, durability, and AI-based analytics for predictive maintenance and advanced surveillance.

6.0 Appendix

a) Project



b) Code Snippet

- **Smart Street Light**

SMART_STREET_LIGHT_.ino

```
#include <WiFi.h>
#include <WiFiClientSecure.h>
#include <UniversalTelegramBot.h>
#include <ArduinoJson.h>
#include "esp_camera.h"

// WiFi credentials
const char* ssid = "ris";                  // Replace with your hotspot/router SSID
const char* password = "nandar2628";       // Replace with your hotspot/router password

// Telegram bot token and chat ID
const char* BOT_TOKEN = "7697063713:AAG1SNrCz2rx1j2OFOyPCvRHE0LiFzjneWo";
const char* CHAT_ID = "1014727422";

// Telegram setup
WiFiClientSecure client;
UniversalTelegramBot bot(BOT_TOKEN, client);

// Track last message sent
unsigned long lastTelegramSent = 0;
const unsigned long telegramCooldown = 5000; // 5 seconds between alerts

// Pin Definitions
#define PIR_PIN 27         // PIR sensor
#define LDR_DO_PIN 4       // LDR digital pin
#define LDR_AO_PIN 39      // LDR analog pin
#define LED1_PIN 5         // LED1
#define LED2_PIN 16        // LED2
#define LED3_PIN 17        // LED3
```

- **ESP32-CAM**

CameraWebServer.ino | app_httpd.cpp | camera_index.h | camera_pins.h | ci.json

```
#include "esp_camera.h"
#include <WiFi.h>
//
// WARNING!!! PSRAM IC required for UXGA resolution and high JPEG quality
//
selected
//
size
//
//
Ensure ESP32 Wrover Module or other board with PSRAM is
Partial images will be transmitted if image exceeds buffer
You must select partition scheme from the board menu that has
at least 3MB APP space.
//
// ==================
// Select camera model
// ==================
//#define CAMERA_MODEL_WROVER_KIT // Has PSRAM
//#define CAMERA_MODEL_ESP_EYE // Has PSRAM
//#define CAMERA_MODEL_ESP32S3_EYE // Has PSRAM
//#define CAMERA_MODEL_M5STACK_PSRAM // Has PSRAM
//#define CAMERA_MODEL_M5STACK_V2_PSRAM // M5Camera version B Has PSRAM
//#define CAMERA_MODEL_M5STACK_WIDE // Has PSRAM
//#define CAMERA_MODEL_M5STACK_ESP32CAM // No PSRAM
//#define CAMERA_MODEL_M5STACK_UNITCAM // No PSRAM
#define CAMERA_MODEL_AI_THINKER // Has PSRAM
//#define CAMERA_MODEL_TTGO_T_JOURNAL // No PSRAM
//#define CAMERA_MODEL_XIAO_ESP32S3 // Has PSRAM
#include "camera_pins.h"
void startCameraServer();
void setupLedFlash(int pin);
void setup() {
Serial.begin(115200);
  Serial.setDebugOutput(true);
```